Lecture 5: Ethics for AI and Robotics

COMP3018/COMP5018: Human-Robot Interaction

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Introduction

- **Today's Topics**

1- Ethics for AI and Robotics

- **Session Learning Outcomes**

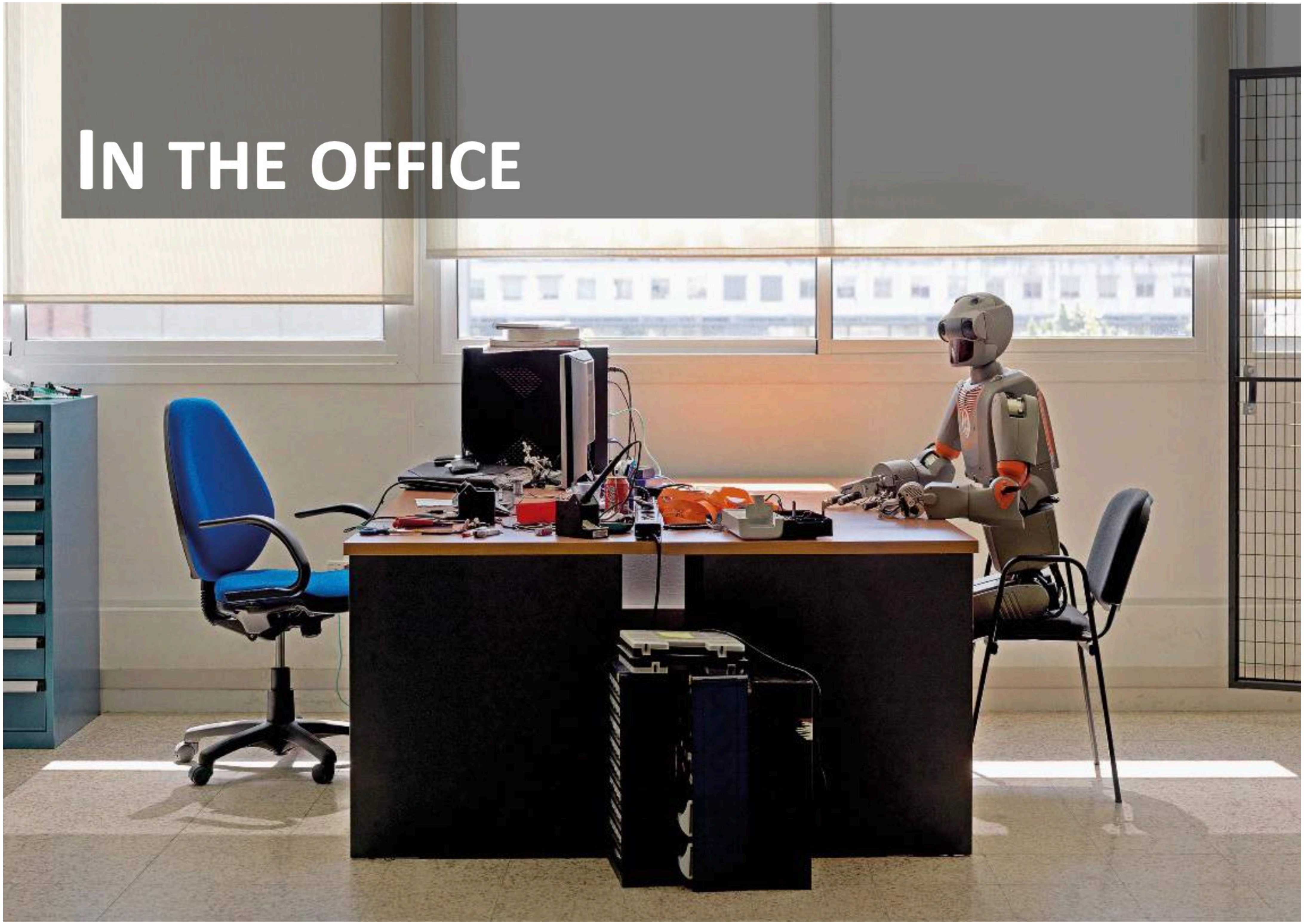
By the end of today's lecture you will be able to:

1- Describe the ethical challenges raised by the use of technology

2- Describe the approaches through which we can consider ethics in AI

Technology and AI Nowadays …

Technology and AI Nowadays …



Technology and AI Nowadays …

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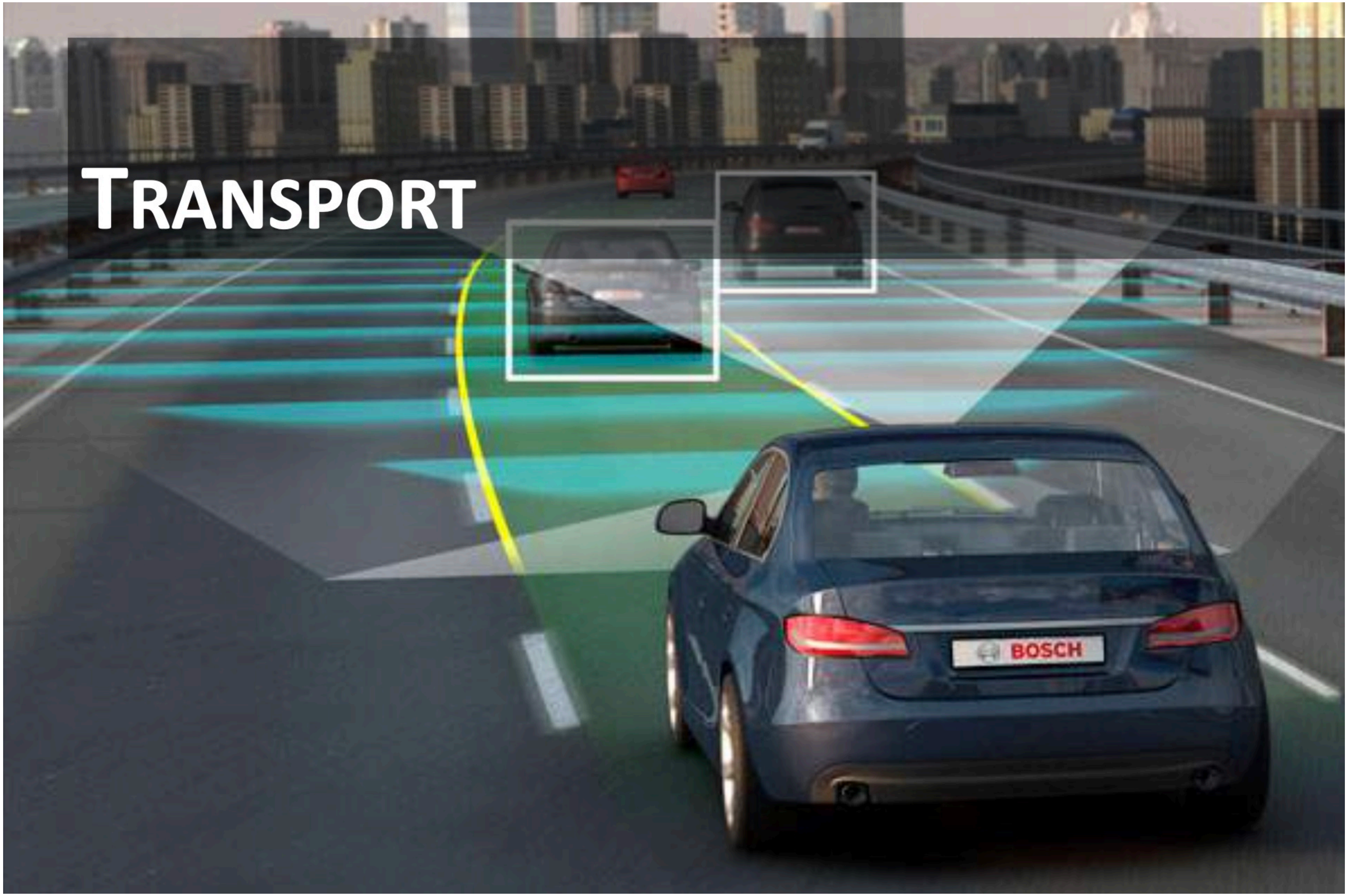


Technology and AI Nowadays …

Technology and AI Nowadays …

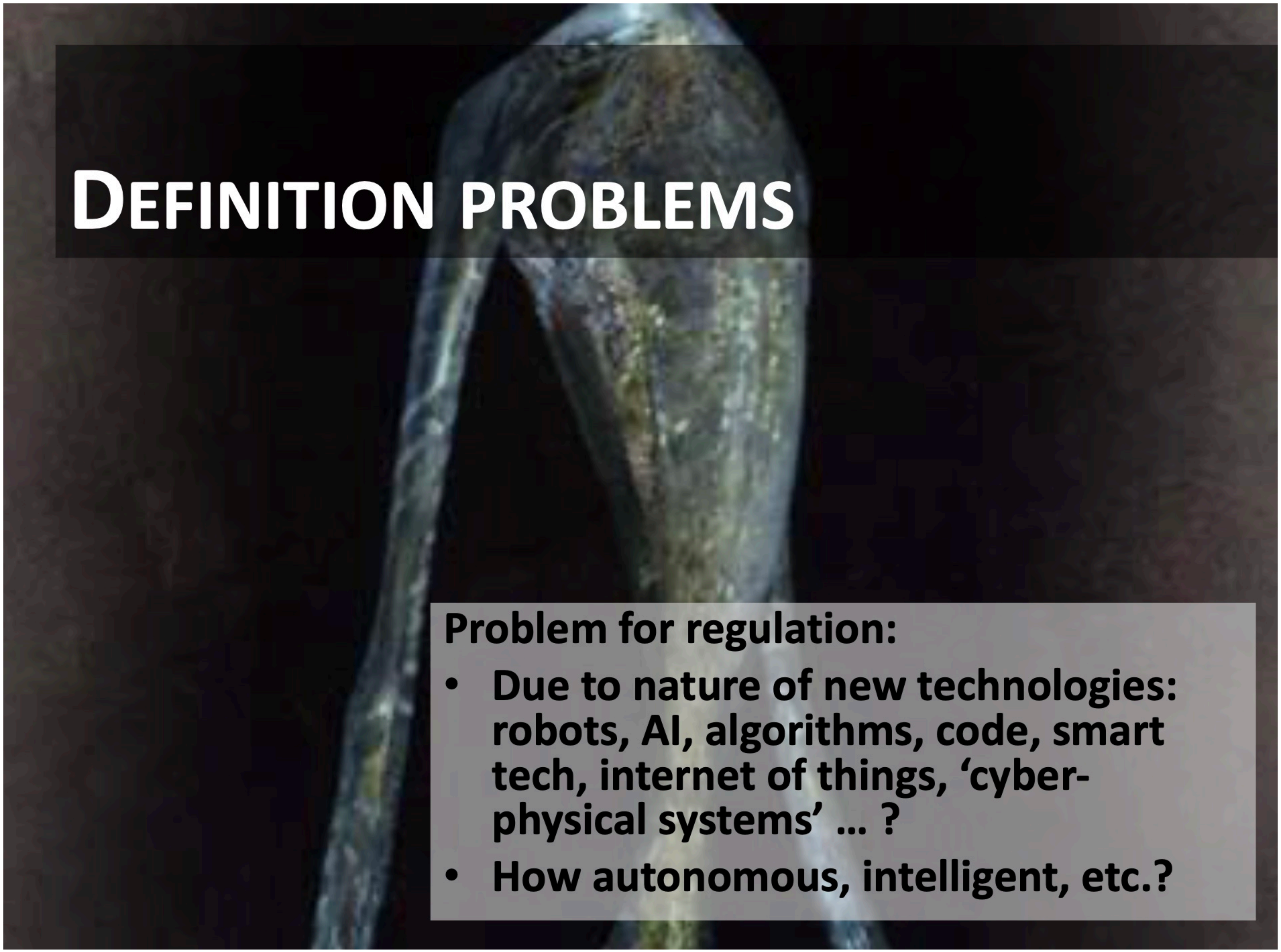
Technology and AI Nowadays …

DATA

Technology and AI Nowadays …

ALL THINGS - EVERYWHERE

Some Issues Raised due to the Use of Technology



Some Issues Raised due to the Use of Technology

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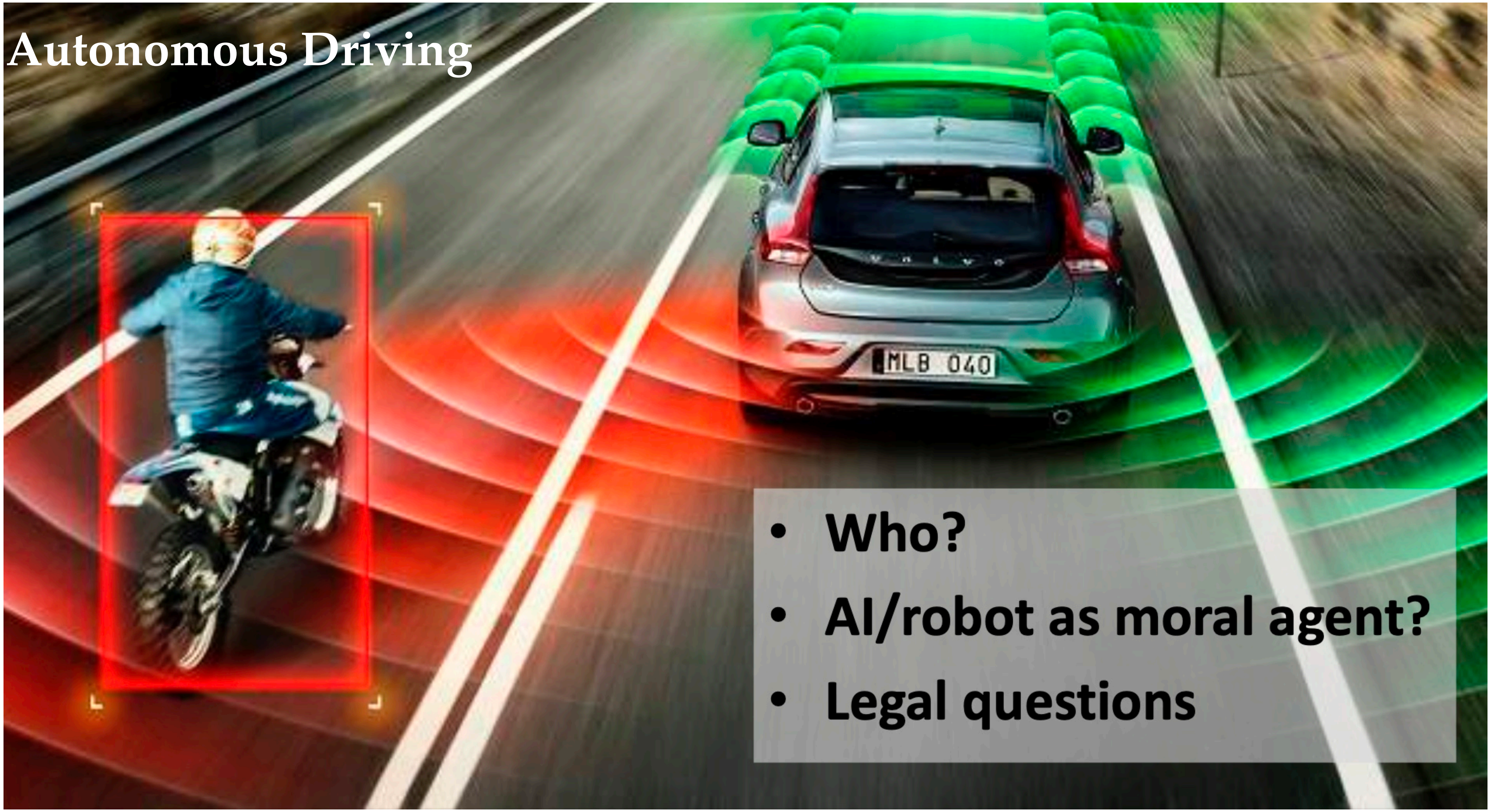


Some Issues Raised due to the Use of Technology

- <u>Moral and Legal Responsibility</u>

<u>Examples</u>

- An autonomous car drives into a group of children
- AI causes crashes in financial markets
- Machines harm workers in a factory
- Care robot gives the wrong medication
- Killer robot kills civilian
- A child gets too attached to an educational robot

Some Issues Raised due to the Use of Technology

Examples

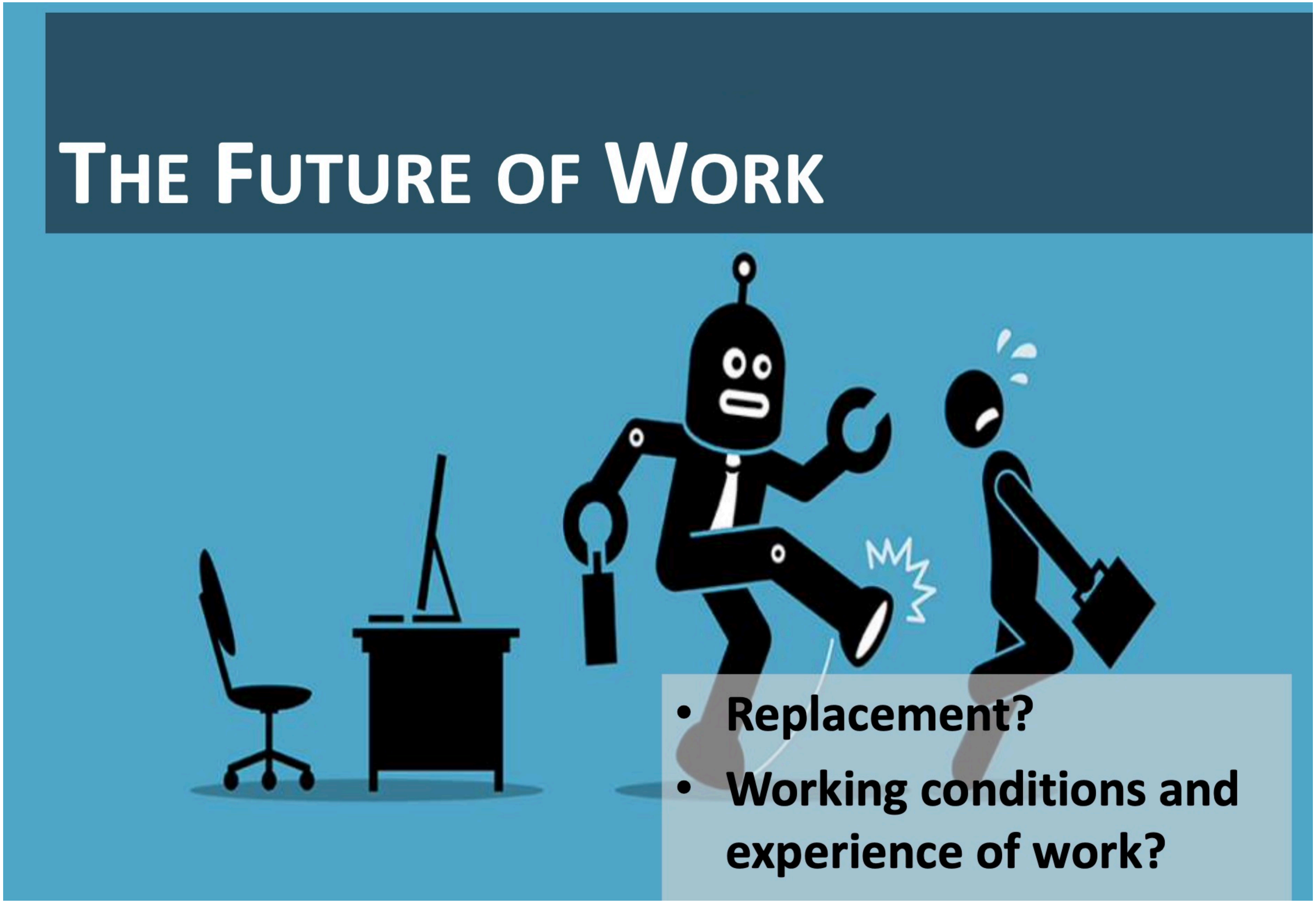
- An autonomous car drives into a group of children

Action: Classification (Society of Automotive Engineers - SAE) / 6 levels of self-driving:

– Level 0: monitoring, warnings
– Level 1: adaptive cruise control, automated parking
– Level 2: automated driving, but the driver must be alert and be able to take over any time
...
– Level 5: no human intervention is needed

But still, not enough, more is needed

Societal Implications of AI



Societal Implications of AI

Technological Forecasting & Social Change 114 (2017) 254–280

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Technological Forecasting & Social Change

ELSEVIER

The future of employment: How susceptible are jobs to computerisation?☆

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ABSTRACT

We examine how susceptible jobs are to computerisation. To assess this, we begin by implementing a novel methodology to estimate the probability of computerisation for 702 detailed occupations, using a Gaussian process classifier. Based on these estimates, we examine expected impacts of future computerisation on US labour market outcomes, with the primary objective of analysing the number of jobs at risk and the relationship between an occupations probability of computerisation, wages and educational attainment.

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European Economic and Social Committee

INT/806
Artificial intelligence

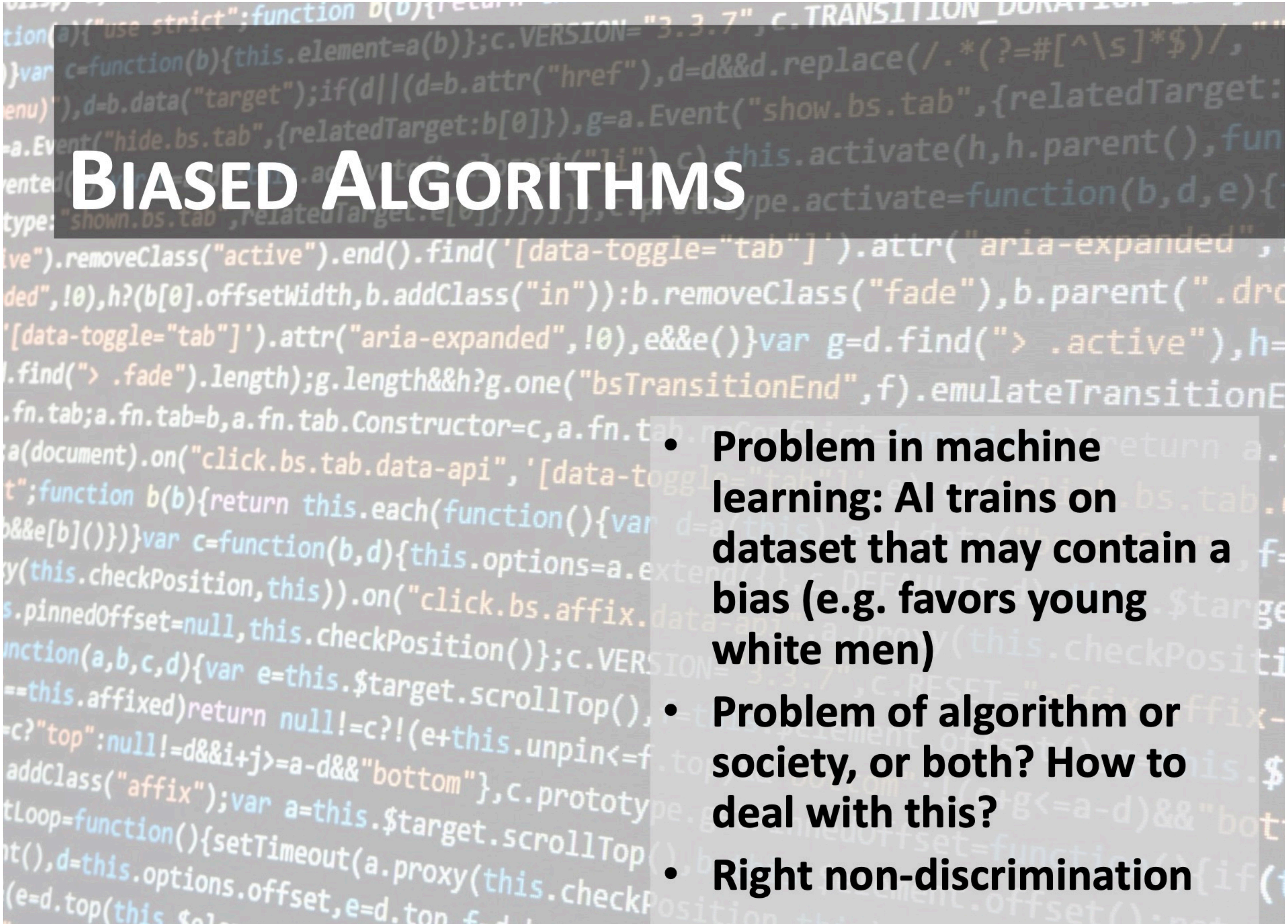
OPINION

Section for the Single Market, Production and Consumption

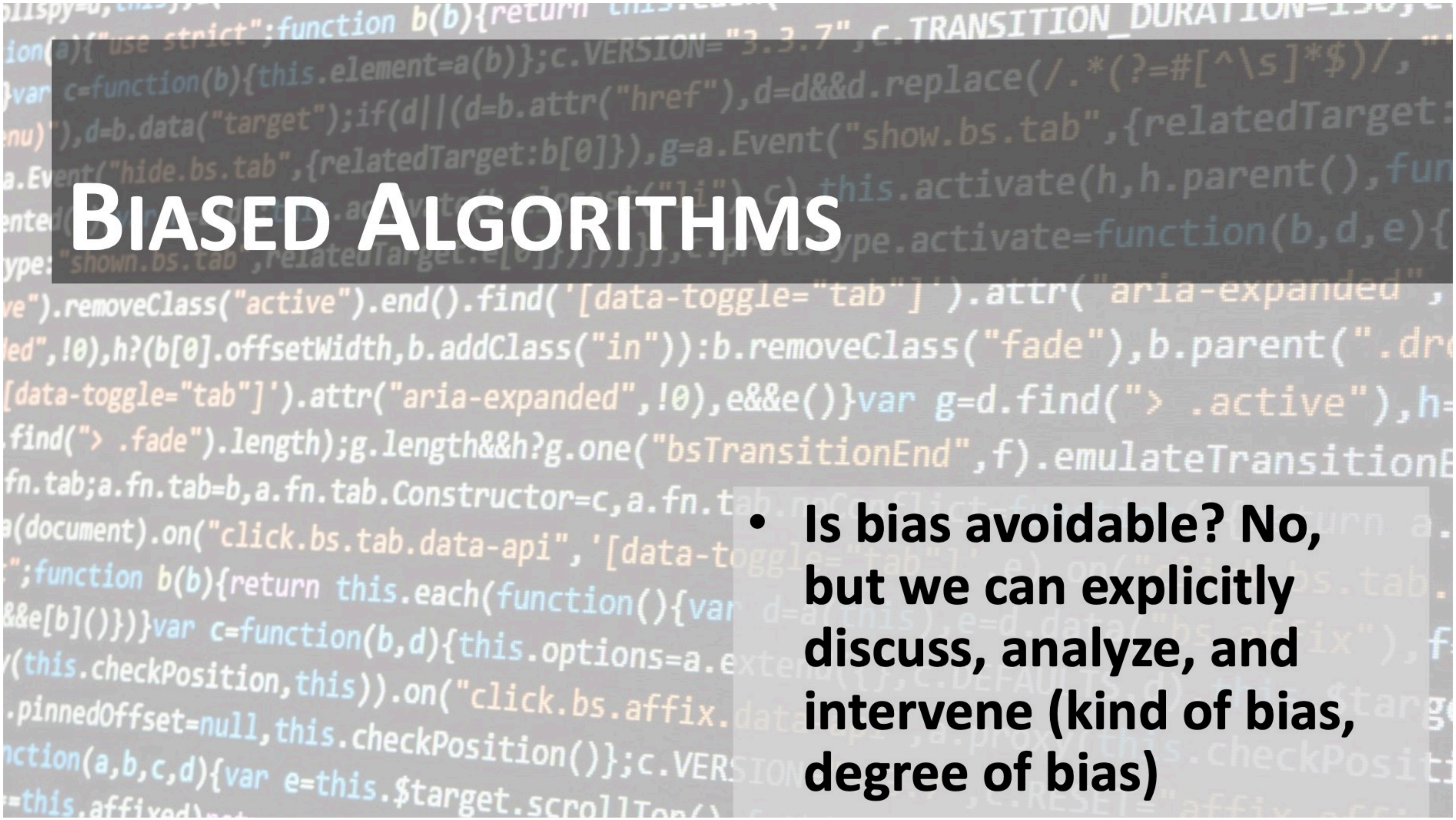
Artificial intelligence – The consequences of artificial intelligence on the (digital) single market, production, consumption, employment and society
(own-initiative opinion)

Rapporteur: Catelijne MULLER

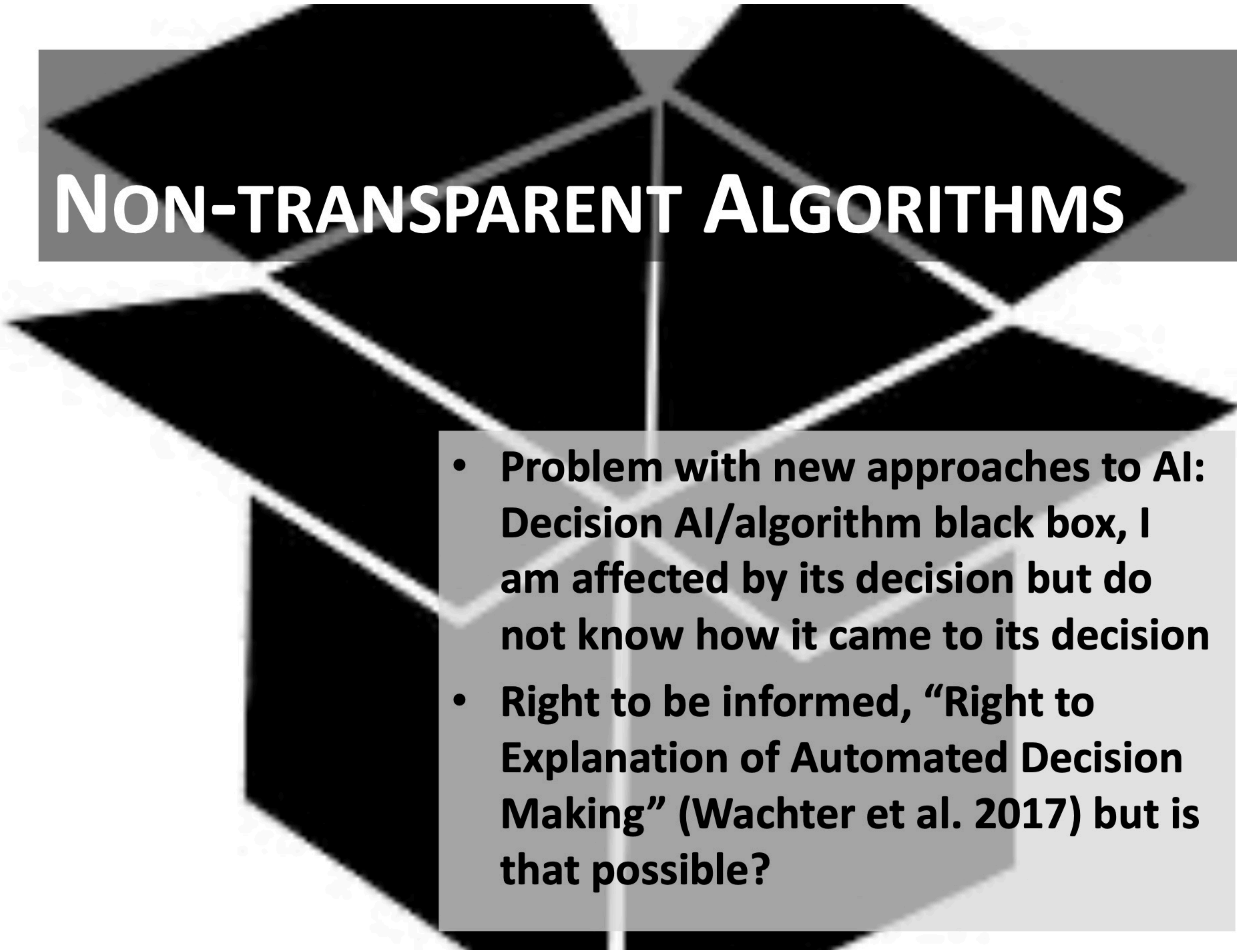
Societal Implications of AI



Societal Implications of AI



Societal Implications of AI



Non-transparent Algorithms

- Problem with new approaches to AI: Decision AI/algorithm black box, I am affected by its decision but do not know how it came to its decision
- Right to be informed, “Right to Explanation of Automated Decision Making” (Wachter et al. 2017) but is that possible?

AI - Ethics

Regulations are needed, legal frameworks are needed, …

Ethics

- **Ethics is a branch of philosophy that deals with moral principles and values that guide human behavior. Ethics is concerned with what is right and wrong, good and bad, and what ought to be done or avoided. In the context of AI and robotics, ethics refers to the principles and values that should guide the *development, deployment, and use of these technologies*.**

- **Meta-Ethics**
  - **Studying where our ethics come from**
  - **Are they derived from human nature, society, religion, cultural norms, or something universal and objective?**

- **Normative Ethics**
  - **Generating moral standards for right vs. wrong**
  - **The consequences of our behaviors on others**

- **Applied Ethics**
  - **Examining specific controversial issues (nuclear war, animal rights)**

Ethics

Some examples of scientific research in which ethics play a large role:

- **Stem cell research**
- **Cloning/genetically modified food**
- **Nuclear technology**
- **Animal rights**
- **Medical trials**
- **Disease research**

- **New technologies have unintended negative side effects**
- **Scientists and engineers must think about:**

- how they should act on the job

- what projects should or should not be done

- and how they should be handled

Considering Ethics in AI: Approaches

Considering Ethics in AI: Approaches

- **Bottom-up approach**

Considering Ethics in AI: Approaches

Considering Ethics in AI: Approaches

Ethics: How not to do it

Thousands of drivers suffer loss of power following VW emissions 'fix'

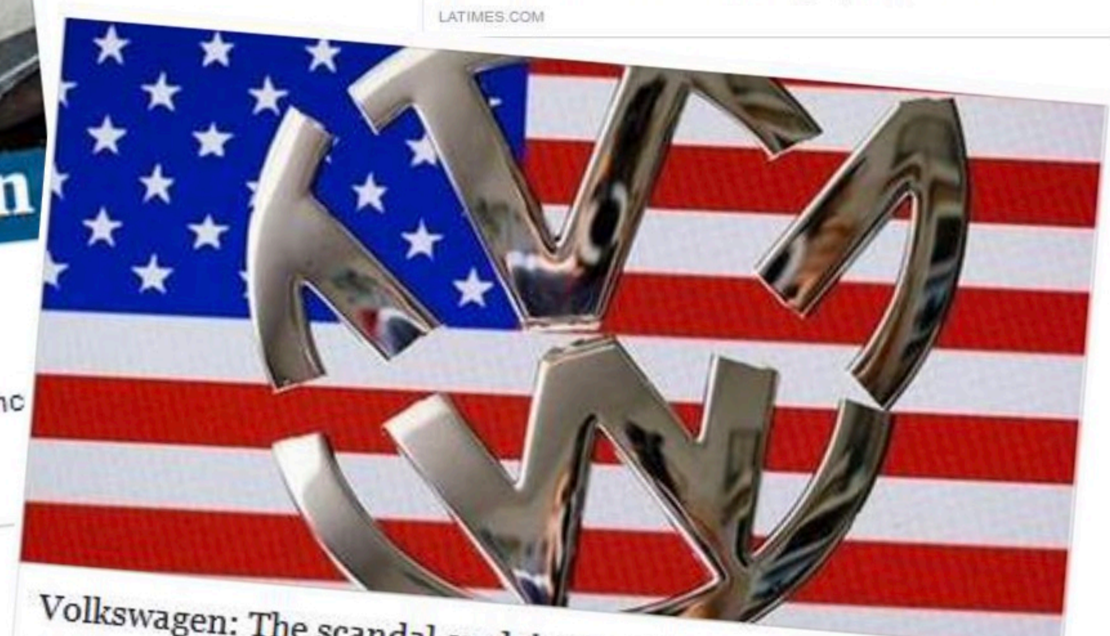
41,000 owners are bringing a class action against the manufacturer citing poor performanc worse fuel consumption – and no compensation

THEGUARDIAN.COM

Volkswagen executive pleads guilty in emissions scandal

A German Volkswagen executive pleaded guilty Friday to conspiracy and fraud charges in Detroit in a scheme to cheat emission rules on nearly 600,000 diesel vehicles.

LATIMES.COM



Volkswagen: The scandal explained - BBC News

The scandal over VW cheating pollution emissions tests in the US is casting a cloud over the whole car industry.

BBC.COM | BY BBC NEWS

Considering Ethics in AI: Approaches

- Ensure that automation designs and deployments do not create ethical hazards for individuals and society.

Considering Ethics in AI: Approaches

Also Non-Governmental Actors!

RESPONSIBLE

ROBOTICS

Accountable Innovation for the Humans Behind the Robots

Ethics in AI


Policy needed
Everyone affected, need for vision and policy NOW

Summary

- **Today we discussed**

1- Ethics for AI and Robotics

- **Next week**

1- We will discuss a different topic; i.e., Models for Human-Robot Collaboration